Vantage Detection API Interface
Version V1.0

Innovating Through Informatics™
For the exclusive use of the VHB.

4936432-A10 Rev B July 2020

© 2020 ITERIS, INC.

All rights reserved. No parts of this work may be reproduced in any form or by any means - graphic, electronic, or mechanical, including photocopying, recording, taping, or information storage and retrieval systems - without the prior written permission of Iteris, Inc.

Products that are referred to in this document may be either trademarks and/or registered trademarks of the respective owners. The publisher and the author make no claim to these trademarks.

While every precaution has been taken in the preparation of this document, the publisher and the author assume no responsibility for errors or omissions, or for damages resulting from the use of information contained in this document or from the use of programs and source code that may accompany it. In no event shall the publisher and the author be liable for any loss of profit or any other commercial damage caused or alleged to have been caused directly or indirectly by this document.

The information contained in this document is subject to change without notice.

TABLE OF CONTENTS

1 Introduction 4
2 Vantage Detection API 5
2.1 General 5
2.2 Definitions 5
2.3 Function Call Definitions 6
2.4 Elements in Data Stream 6
3 API: Count Data 7
4 API: Zone Status 8
5 API: Channel Output Status 9
6 API: ALL Data Streams 10

TABLES

Table 1 – List of Data Streams 5
Table 2 – List of API Function Calls 6
Table 3 – List of Elements in Count Stream 6
Table 4 – List of Elements in Zone Status Stream 6
Table 5 – List of Elements in Channel Status Stream 6
Table 6 – List of Count Zone Types 7

For the exclusive use of the VHB.

1 INTRODUCTION

The Vantage Detection API provides real time data streams from Iteris detection platforms. It uses a message broker that implements the MQTT protocol version 5.0.

It is lightweight and is suitable for use on all devices from low power single board computers to full servers.

The MQTT protocol provides a lightweight method of carrying out messaging using a publish/subscribe model. This makes it suitable for Internet of Things messaging with sensors or remote devices such as embedded computers or microcontrollers.

A MQTT client is necessary to subscribe to the stream and receive data from a detection system. The client is recommended to be on MQTT protocol version 5.0. At a minimum the client should be protocol version 3.1.1.

Additional information can be found in the user guides for the various Vantage detection platforms.

For the exclusive use of the VHB.

2 VANTAGE DETECTION API

2.1 General

The Vantage Detection API can provide the following data streams:

Table 1 – List of Data Streams

DATA STREAM	DESCRIPTION
Count	Provides data for all available counts in the system. Updated every 60 seconds.
Zone Status	Provides status for all configured zones in the system. Updated every 100mS.
Channel Status	Provides status for all configured channel outputs in the system. Updated every 100mS.

Data stream is in JSON format. Samples of the output can be found in the sections below.

2.2 Definitions

The request in broken into three sections:

Header
This is where the function call is defined. See section below.

Body
This is a list of the variable items that may be added to a function call. Currently the only item to define is the IP address of the target system.

Footer
This provides various information to support the data broker on the target system.

Sample request

mosquitto_sub -t /iteris/count/ -h 192.168.1.20 --cafile ca.crt --cert client.crt --key client.key

Header	Body	Footer
mosquitto_sub -t /iteris/count/ -h	192.168.1.20	--cafile ca.crt --cert client.crt --key client.key

For the exclusive use of the VHB.

2.3 Function Call Definitions

This is a list of the current API calls available from the system.

Table 2 – List of API Function Calls

FUNCTION	DESCRIPTION
mosquitto_sub -t /iteris/count/	Subscribe to count message
mosquitto_sub -t /iteris/zone/	Subscribe to zone status message
mosquitto_sub -t /iteris/channel/	Subscribe to channel output status message
mosquitto_sub -t /iteris/#	Subscribe to all 3 messages

2.4 Elements in Data Stream

Table 3 – List of Elements in Count Stream

ELEMENT	DESCRIPTION
“date”	Date and time in ISO time format
“z#”	Zone number
data	Total count recorded for the zone in the last 60 seconds

Table 4 – List of Elements in Zone Status Stream

ELEMENT	DESCRIPTION
“type”	“zone” indicates zone status data
“z#”	Zone number
status	0 indicated zone is inactive (not detecting) 1 indicates zone is active (detecting)

Table 5 – List of Elements in Channel Status Stream

ELEMENT	DESCRIPTION
“type”	“channel” indicates channel output status data
“z#”	Zone number
status	0 indicated channel is inactive (no output) 1 indicates zone is active (output)

3 API: COUNT DATA

The data stream provides data for all available counts in the system. The steam is updated every 60 seconds.

Subscribe String

mosquitto_sub -t /iteris/count/ -h 192.168.1.20 --cafile ca.crt --cert client.crt --key client.key

Table 6 – List of Count Zone Types

NUMBER RANGE	DESCRIPTION
1-128	Manually configured video vehicle and bicycle count zones
R1-R128	Manually configured radar vehicle count zones
601-608	PedTrax pedestrian count zones
651-654	Vector Tripline count zones
701-740	Automatic vehicle counts by lane

Sample Result


```
{
"date":"2020-06-23T12:18:00-0800",
"Z1":4,"Z2":7,"Z13":10,"Z34":40,"Z45":11,"Z46":34,"Z67":10,"Z78":4,"Z92":31,"Z101":0
,"Z103":23,"Z601":4,"Z602":7,"Z603":0,"Z604":0,"Z605":14,"Z606":0,"Z607":0,"Z608":0,
"Z652":31,"Z701":10,"Z703":0,"Z704":0,"Z705":0,"Z706":100,"Z707":0,"Z708":0,"Z709":0
,"Z711":0,"Z713":4,"Z714":0,"Z715":0,"Z716":0,"Z717":60,"Z718":0,"Z719":0,"Z721":0,"
Z723":0,"Z724":0,"Z725":0,"Z726":0,"Z727":0,"Z728":0,"Z729":0,"Z731":0,"Z733":0,"Z73
4":0,"Z735":0,"Z736":0,"Z737":0,"Z738":0,"Z739":0
}
```

4 API: ZONE STATUS

Provides status for all configured zones in the system. The stream is updated every 100mS.

Subscribe String

mosquitto_sub -t /iteris/zone/ -h 192.168.1.20 --cafile ca.crt --cert client.crt --key client.key

Sample Result


```
{
"type":"zone","Z1":1,"Z2":1,"Z3":1,"Z4":1,"Z5":1,"Z6":1,"Z7":1,"Z8":1,"Z33":0,"Z34":
0,"Z35":0,"Z36":1,"Z37":1,"Z38":0,"Z39":0,"Z40":0,"Z41":0,"Z601":0,"Z602":0,"Z603":0
,"Z604":0,"Z605":0,"Z606":0,"Z607":0,"Z608":0,"Z652":0,"R17":0,"R18":1,"R19":0
}
```

5 API: CHANNEL OUTPUT STATUS

Provides status for all configured channel outputs in the system. This stream is updated every 100mS.

Subscribe String

mosquitto_sub -t /iteris/channel/ -h 192.168.1.20 --cafile ca.crt --cert client.crt --key client.key

Sample Result


```
{
"type":"channel","TS-2:1":1,"TS-2:2":1,"TS-2:4":0,"TS-2:5":1,"TS-2:6":0,"TS-
2:10":0,"TS-2:11":0,"TS-2:12":1,"TS-2:13":0,"TS-2:33":0,"TS-2:34":0,"TS-
2:35":0,"TS-2:36":0,"TS-2:37":0,"TS-2:38":0,"TS-2:39":0
}
```

6 API: ALL DATA STREAMS

Provides subscription to all available data streams from the target system.

Subscribe String

mosquitto_sub -t /iteris/# -h 192.168.1.20 --cafile ca.crt --cert client.crt --key client.key

1700 Carnegie Avenue, Suite 100
Santa Ana, CA 92705

iteris.com

© 2020 Iteris, Inc. All rights reserved.

Innovating Through Informatics™

For the exclusive use of the VHB.